

Financial Analytics

Group Presentation Guidelines

Objective

The goal of this group presentation is to assess your ability to **analyze financial market data**, **quantify financial risk**, and **interpret results from a financial perspective** using the tools introduced in the course.

This is the **only group presentation** for the course.

Presentation Details

- **Duration:** 8 minutes (strict).
- **Format:** In-class presentation.
- **Groups:** Assigned at random.
- **Data:**
 - Main analysis based on historical data from **2023–2025**.
 - Optional: simple evaluation using data from the **first trimester of 2026**.

No Jupyter notebook submission is required. Only presentation slides must be submitted.

Assigned Financial Indexes

Each student is assigned one financial index. The group presentation must jointly analyze the indexes assigned to all group members.

The focus should be on understanding similarities and differences between these indexes from a **financial and risk-based perspective**.

Required Structure of the Presentation

Your presentation must be structured according to the following components.

1. Financial and Business Context

Provide a brief introduction to the indexes analyzed, focusing on:

- The nature of the underlying business or technology.
- The economic sector(s) represented.
- Key structural differences between the indexes (if any).

This section should be concise and serve as context for the quantitative analysis.

2. Exploratory Data Analysis

Perform and present an exploratory analysis of returns, including:

- Time-series plots of prices and returns.
- Summary statistics of returns.
- Visual inspection of volatility and extreme movements.

The goal is to highlight relevant empirical features of the data.

3. Univariate Modeling: Volatility and Risk

For each index, perform a univariate analysis focused on volatility and financial risk:

- Volatility estimation (e.g. rolling volatility, GARCH-type models).

- Quantification of risk using measures discussed in class (e.g. volatility-based metrics, Value-at-Risk, Expected Shortfall).

You must clearly explain:

- what is being measured,
- why it is relevant for financial risk,
- how risk differs across indexes.

4. Portfolio Analysis

Construct a portfolio using the indexes assigned to the group:

- Perform a basic portfolio optimization analysis.
- Discuss diversification effects and risk–return trade-offs.
- Compare portfolio risk with the risk of individual indexes.

The emphasis should be on interpretation, not on technical complexity.

5. Discussion and Interpretation

Conclude the presentation by:

- Summarizing the main findings.
- Highlighting similarities and differences across indexes.
- Discussing financial implications of the results.

Focus on **financial insight**, not methodological detail.

Important Notes

- Clarity and correctness are more important than technical sophistication.
- Slides must be readable, well-organized, and within the time limit.
- Follow the structure above; deviations may be penalized.
- Detailed grading criteria are provided in the rubric available on Blackboard.

Assessment

The presentation will be evaluated according to the official grading rubric provided on Blackboard. The rubric assesses:

- Comprehensiveness of the analysis.
- Correctness of calculations and models.
- Quality of interpretation and financial insight.
- Structure, clarity, and delivery.